US 20250270854A1

(19) United States
(12) Patent Application Publication
Sangster, JR. et al.
(10) Pub. No.: US 2025/0270854 A1
(43) Pub. Date: Aug. 28, 2025

(54) **EXIT DEVICE WITH LOCKDOWN FUNCTION**

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(21) Appl. No.: **19/037,687**

(22) Filed: **Jan. 27, 2025**

Related U.S. Application Data

(60) Provisional application No. 63/557,363, filed on Feb. 23, 2024.

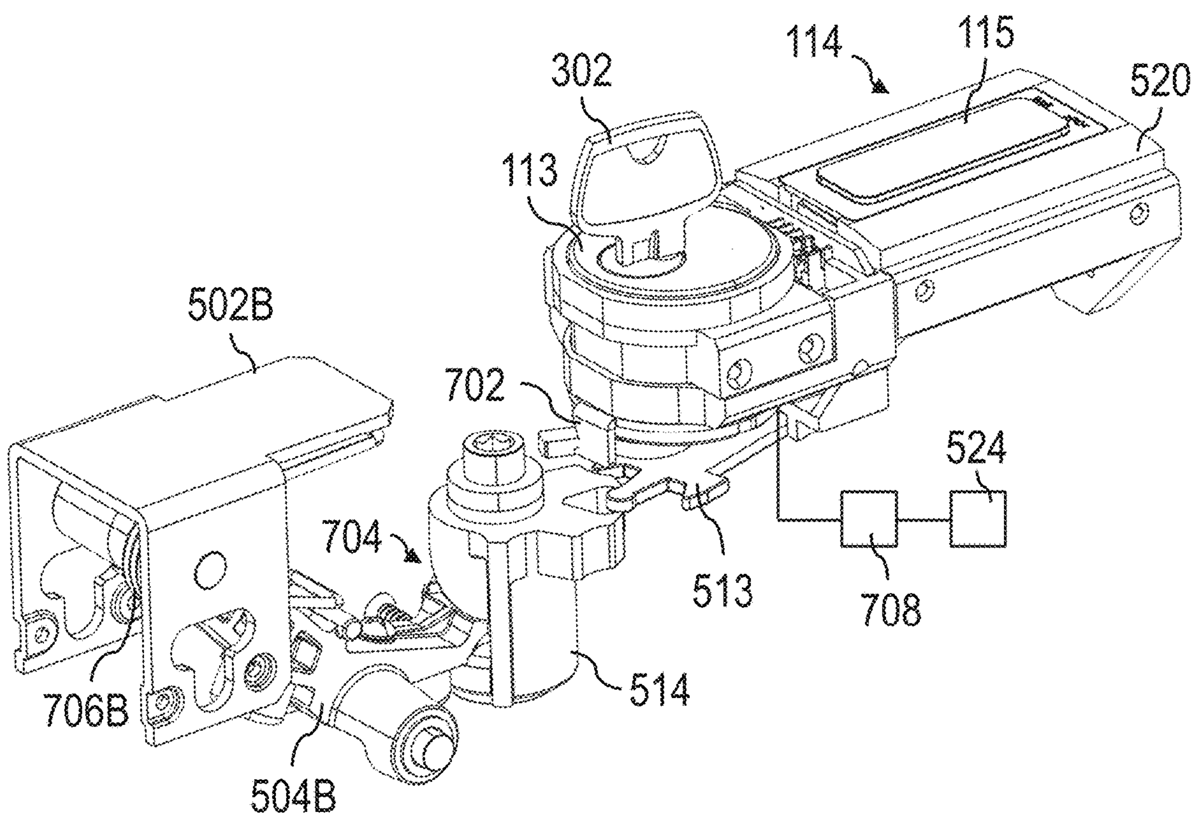
Publication Classification

(51) **Int. Cl.**
E05B 65/10 (2006.01)
E05B 41/00 (2006.01)
E05B 47/00 (2006.01)

(52) **U.S. Cl.**
CPC ***E05B 65/1053*** (2013.01); ***E05B 41/00*** (2013.01); ***E05B 47/0012*** (2013.01); ***E05B 65/108*** (2013.01); *E05B 2047/0017* (2013.01)

(57) **ABSTRACT**

Exit devices with lockdown functionality are disclosed. The exit device may include an indicator configured to indicate a state of the exit device, such as a lockdown state or a non-lockdown state. In the lockdown state, a push bar of the exit device may be prevented from moving. The exit device can include a latch movable between a latch retracted position and a latch extended position. Preventing movement of the push bar may prevent movement of the latch.



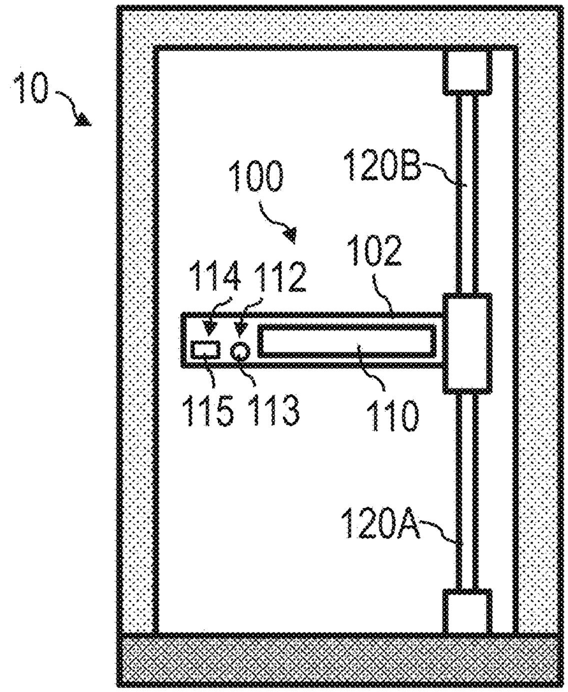


FIG. 1A

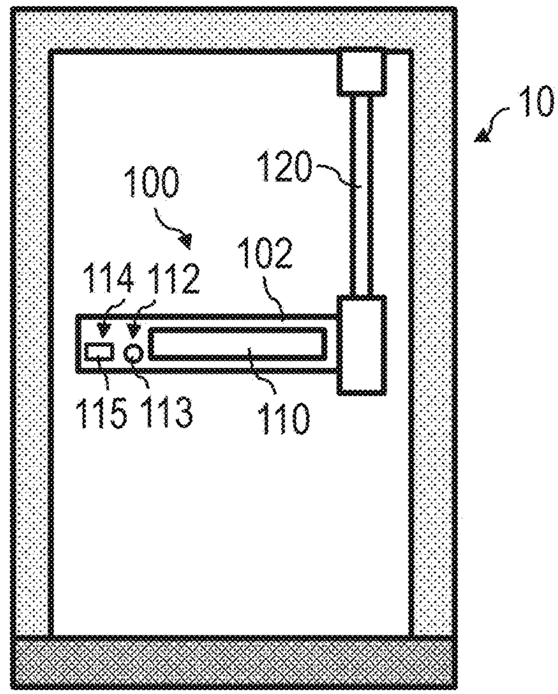


FIG. 1B

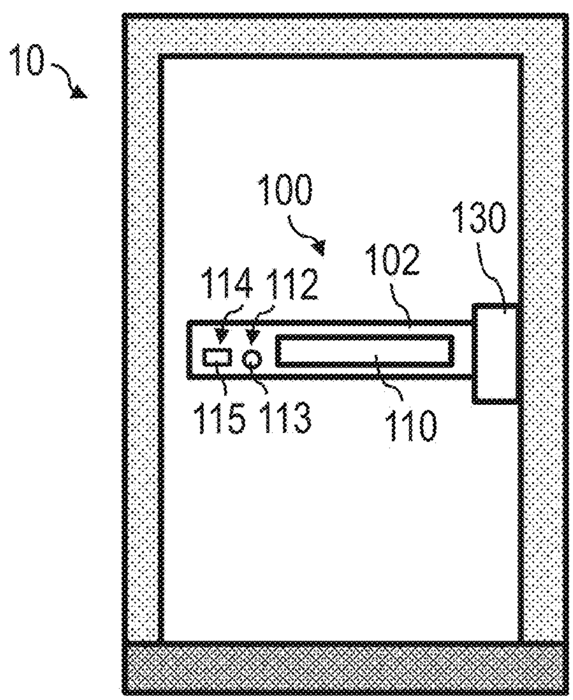


FIG. 2

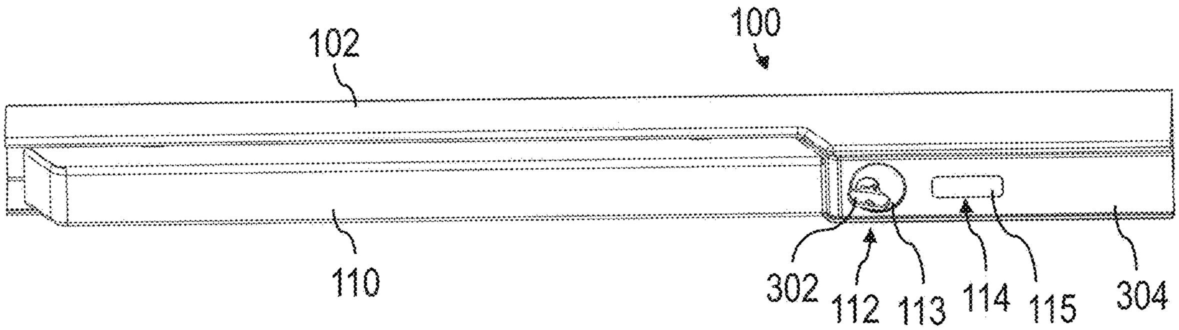


FIG. 3

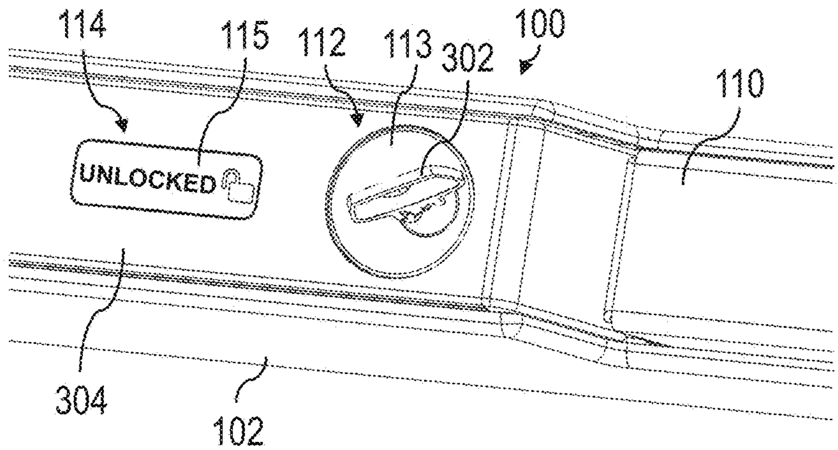


FIG. 4A

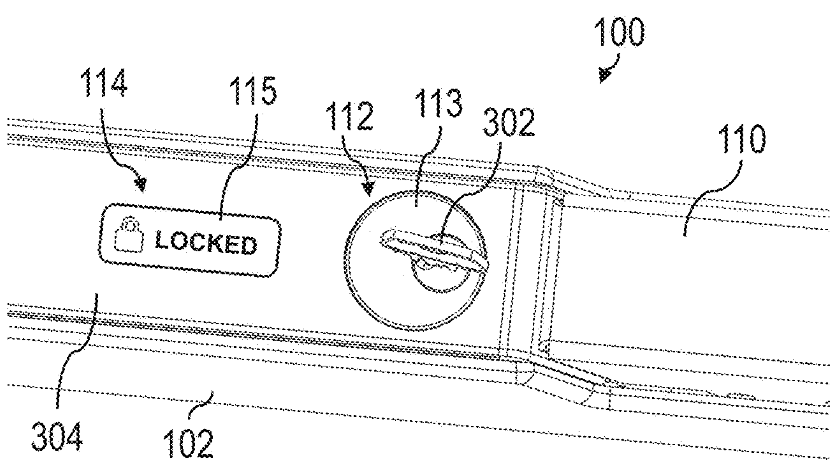
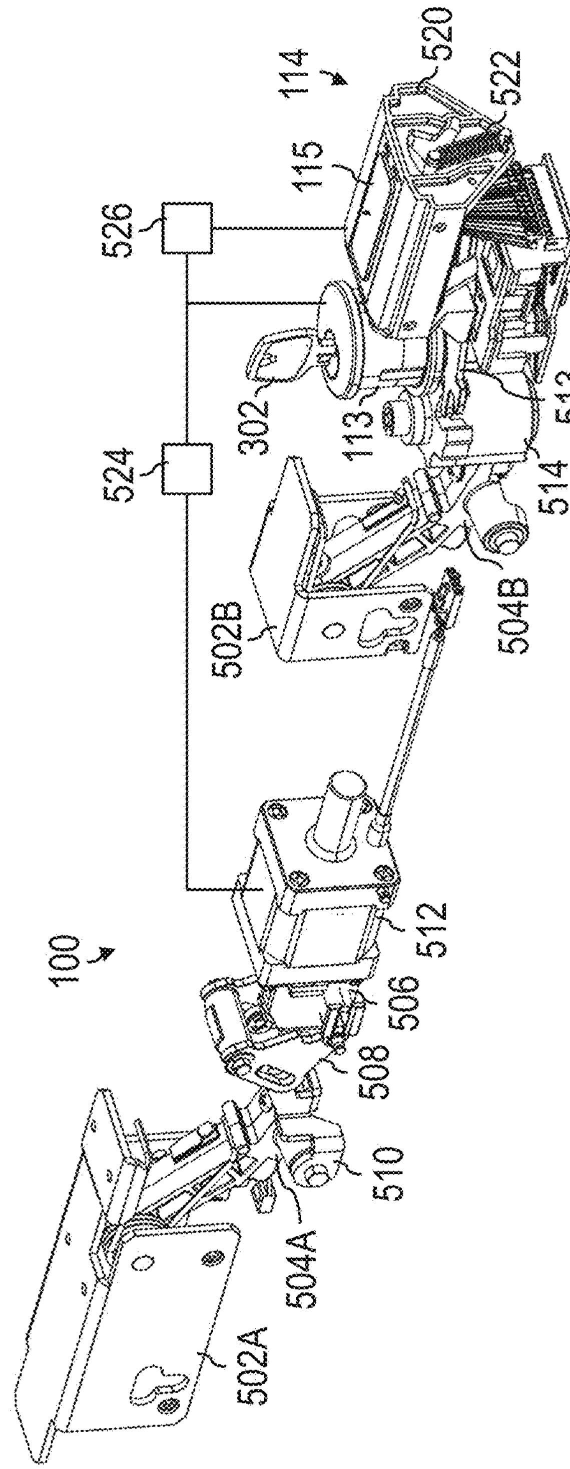
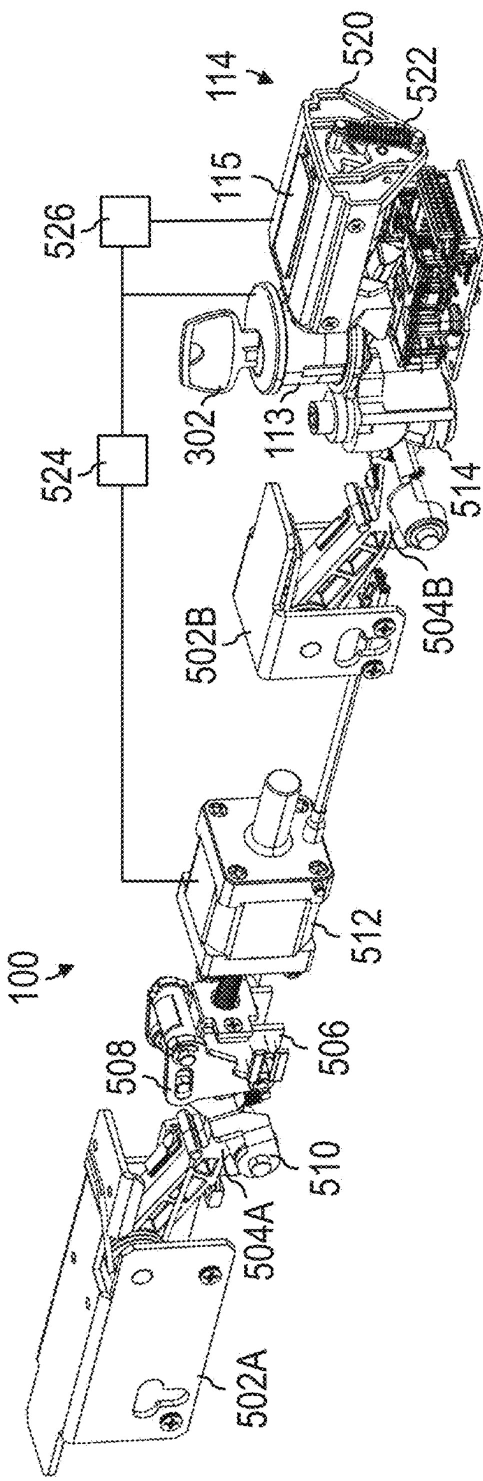


FIG. 4B



100
502A
504A
510
508
506
512
502B
504B
524
514
302
113
513
526
115
114
520
522

FIG. 5A



100
502A
504A
510
508
506
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502B
504B
524
514
302
113
526
115
114
520
522

FIG. 5B

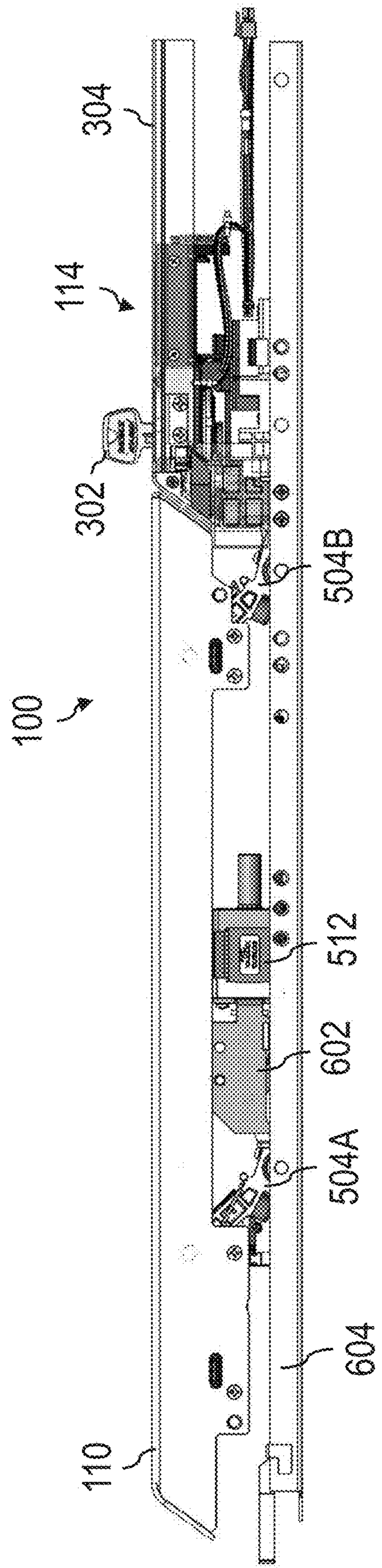
304
114
302
100
504B
512
602
504A
604
110

FIG. 6A

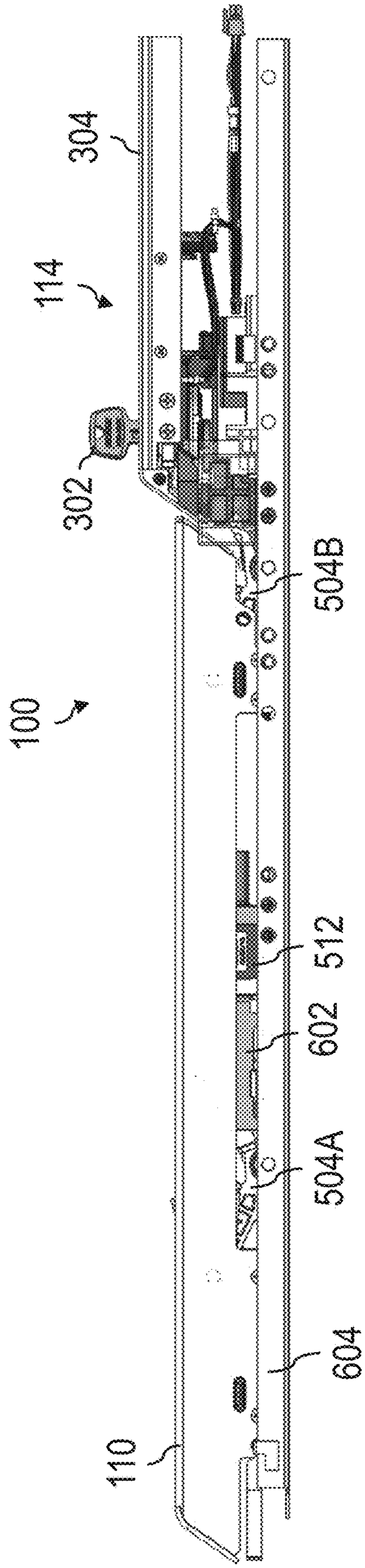
304
114
302
100
504B
512
602
504A
604
110

FIG. 6B

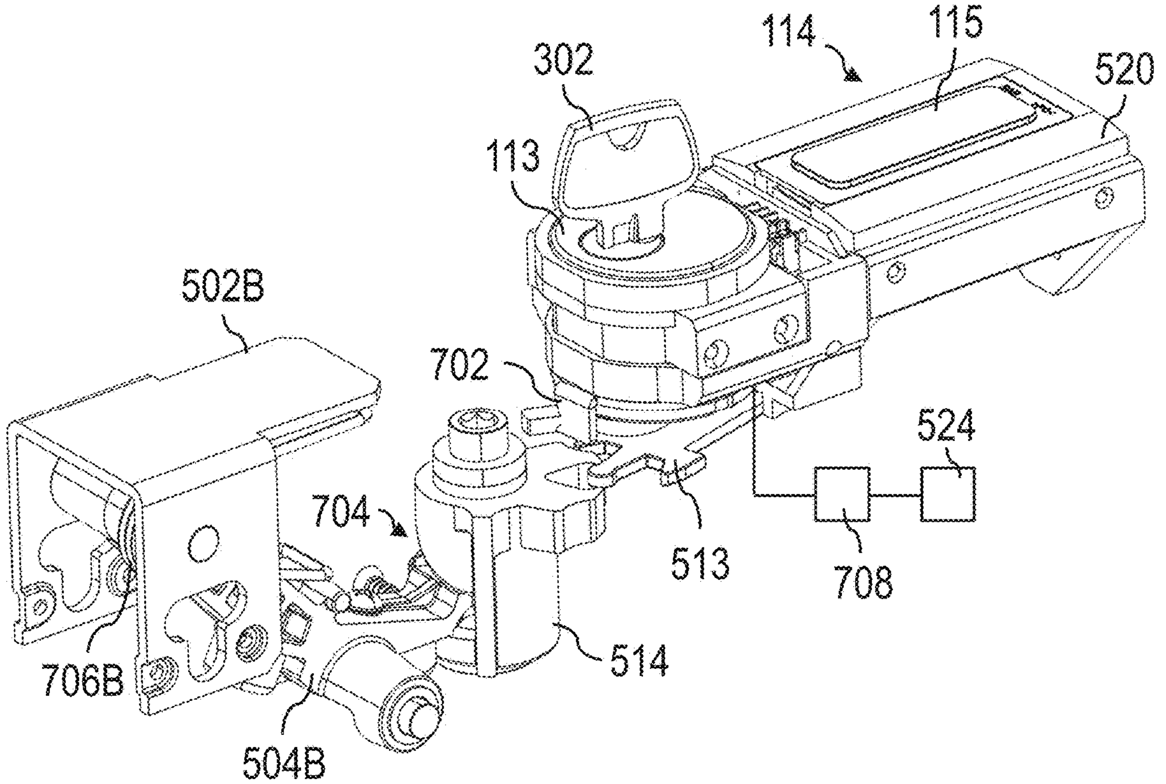
114
115
302
520
113
502B
702
524
704
513
708
706B
514
504B

FIG. 7A

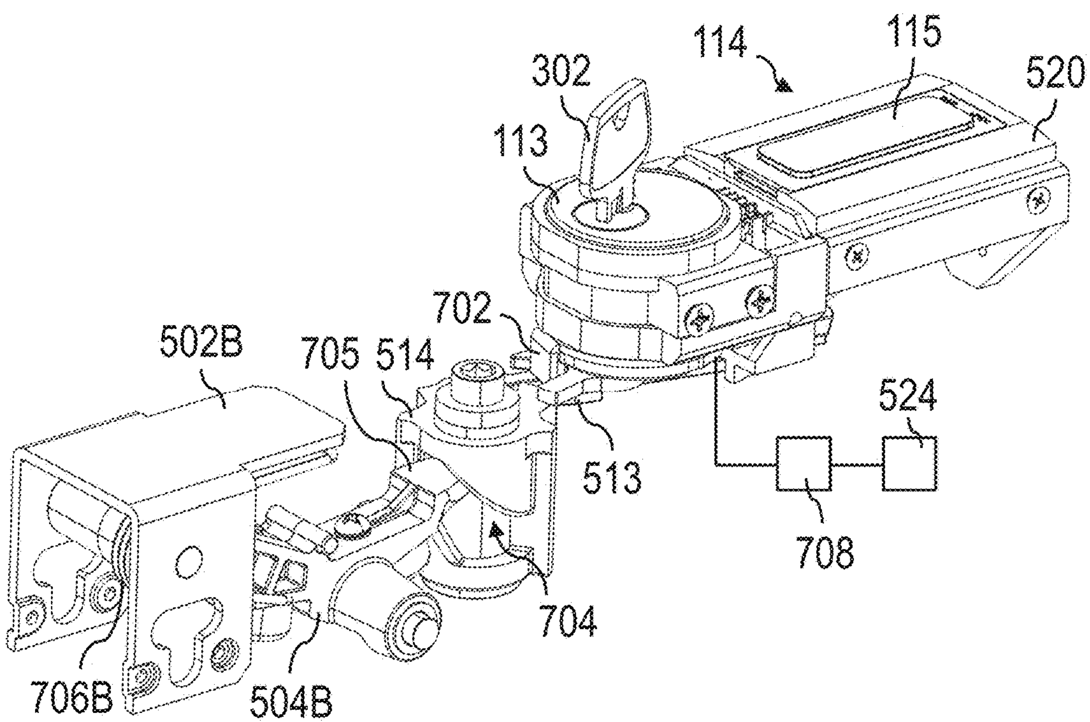
114
115
302
520
113
702
502B
705
514
524
513
708
704
706B
504B

FIG. 7B

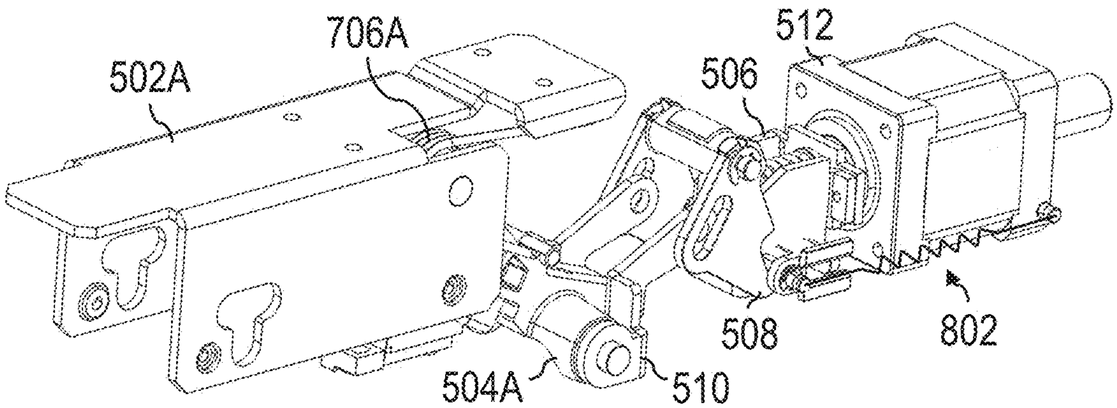

FIG. 8A

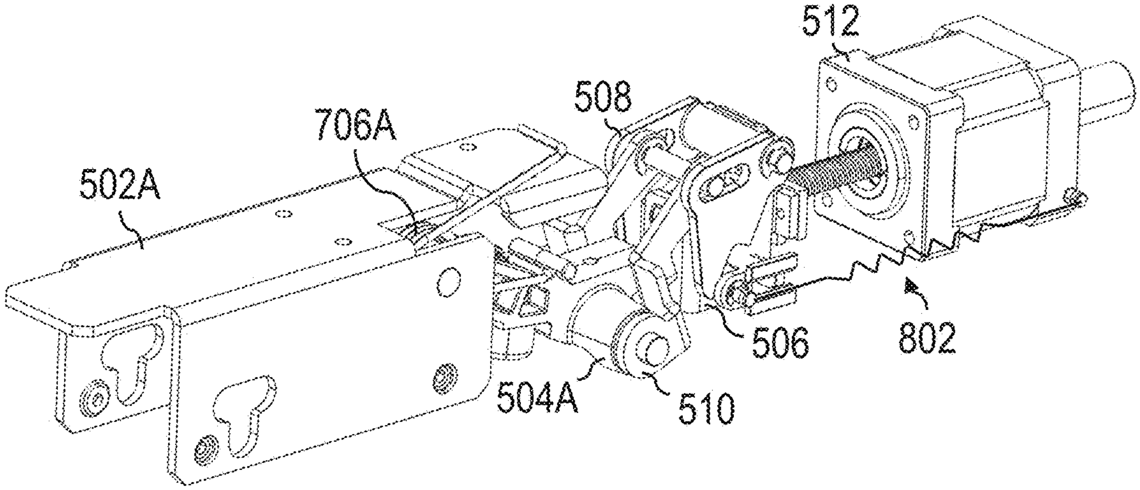

FIG. 8B

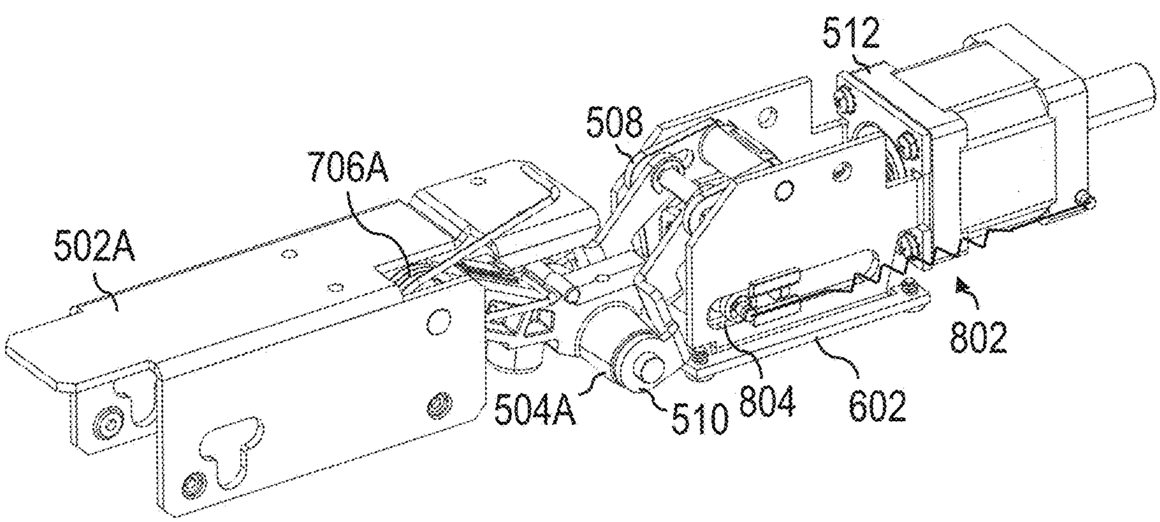

FIG. 8C

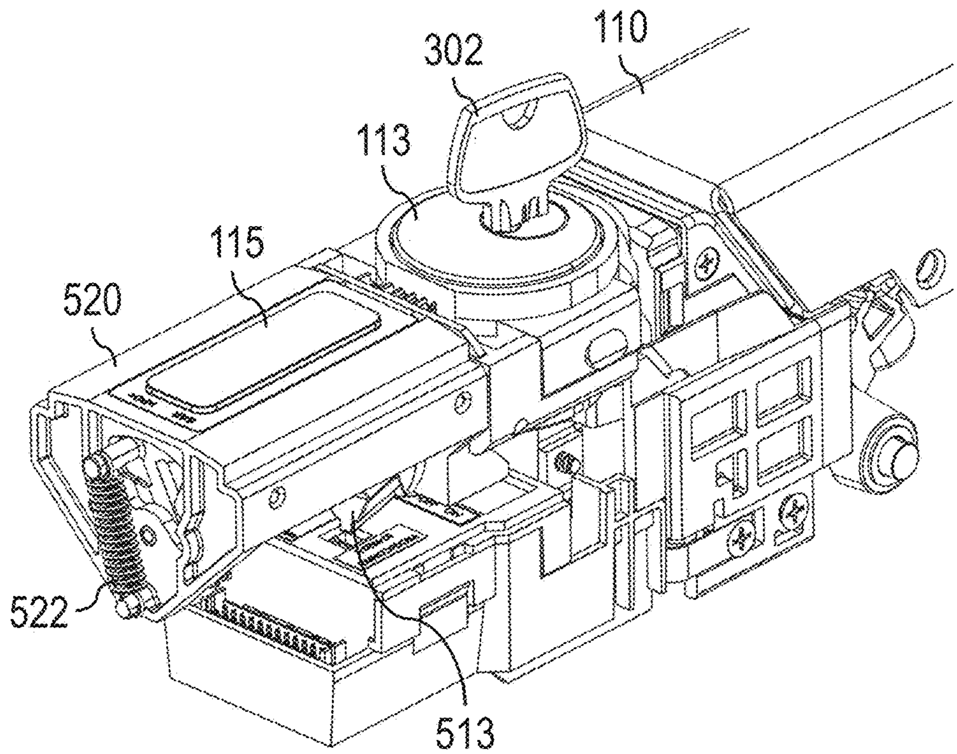

FIG. 9A

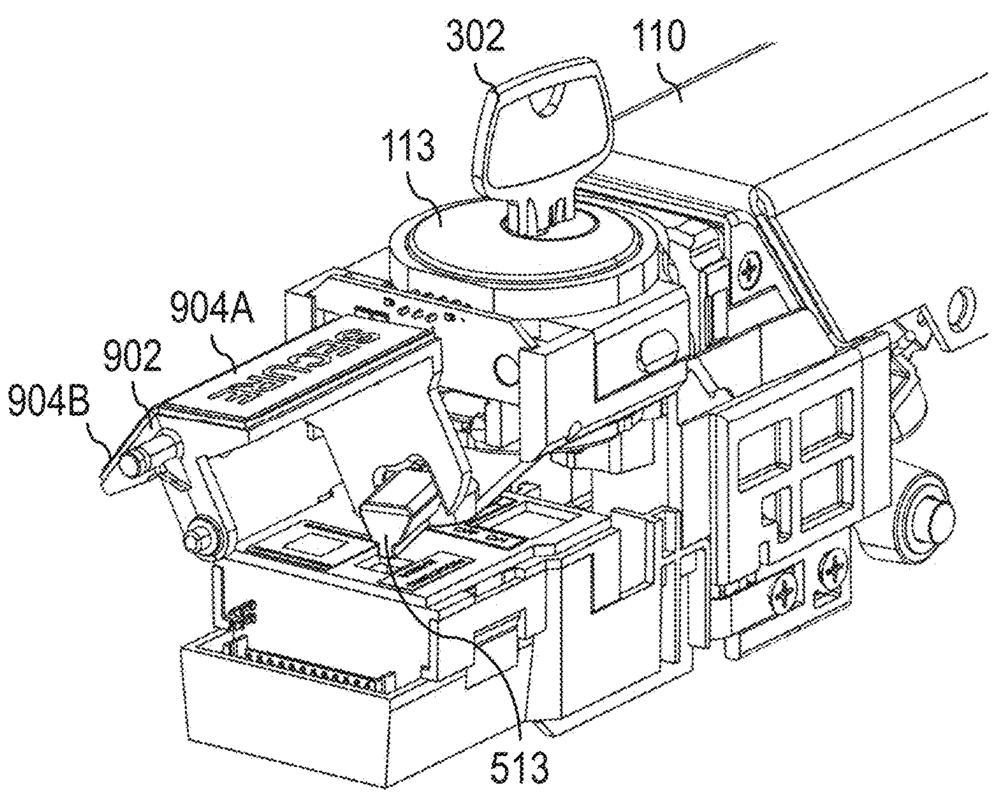

FIG. 9B

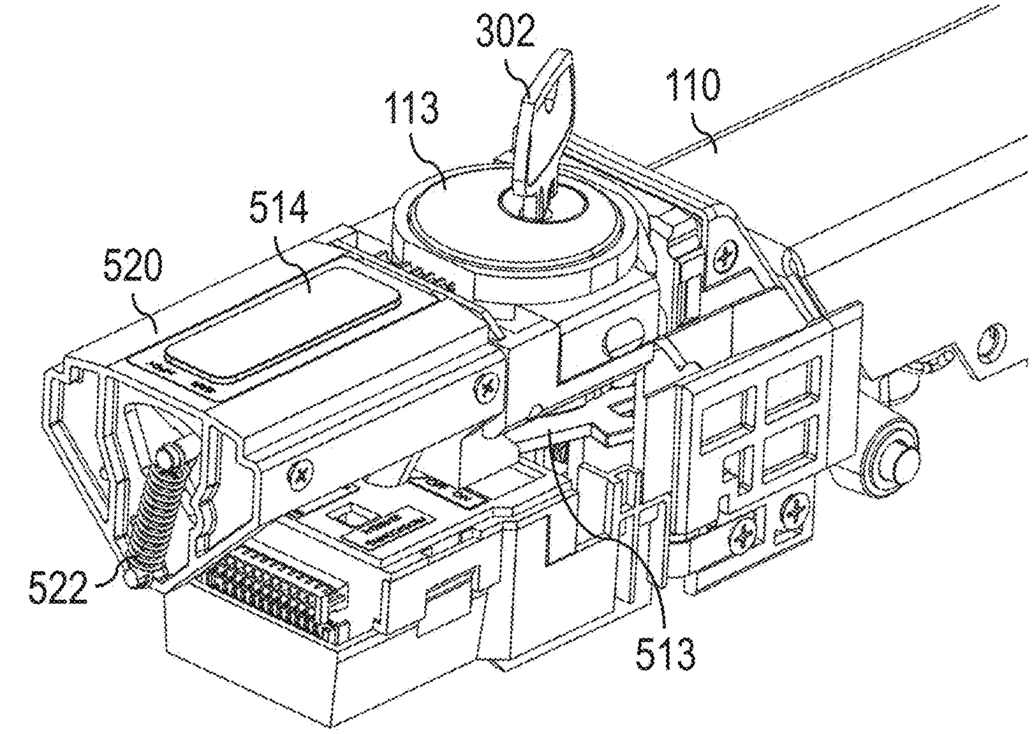

FIG. 10A

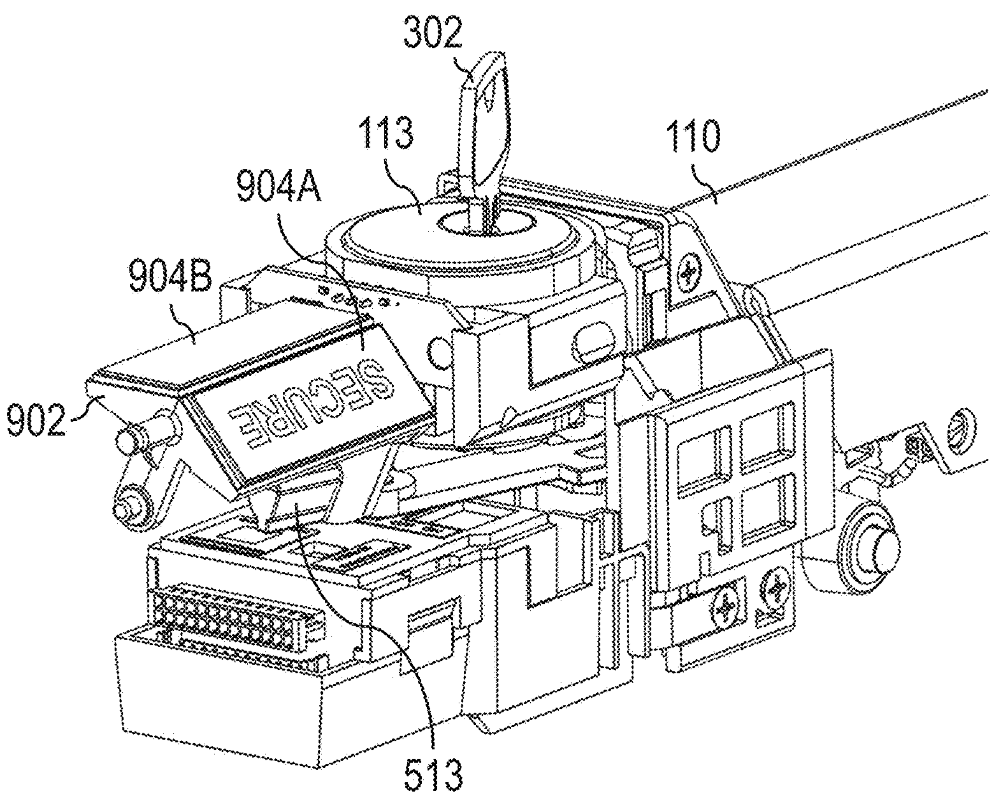

FIG. 10B

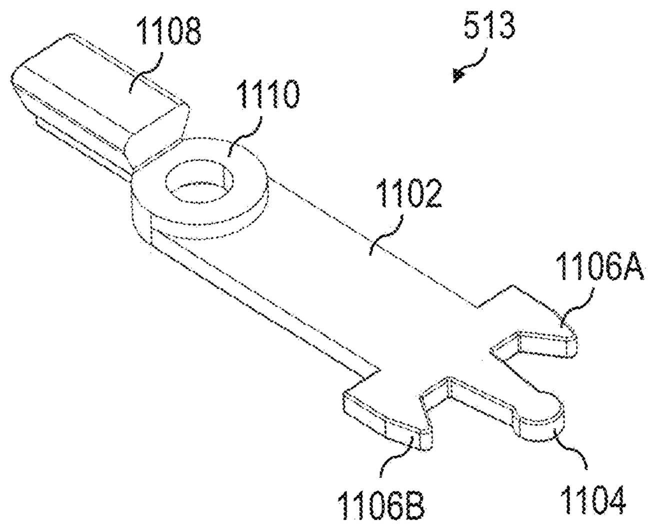

FIG. 11

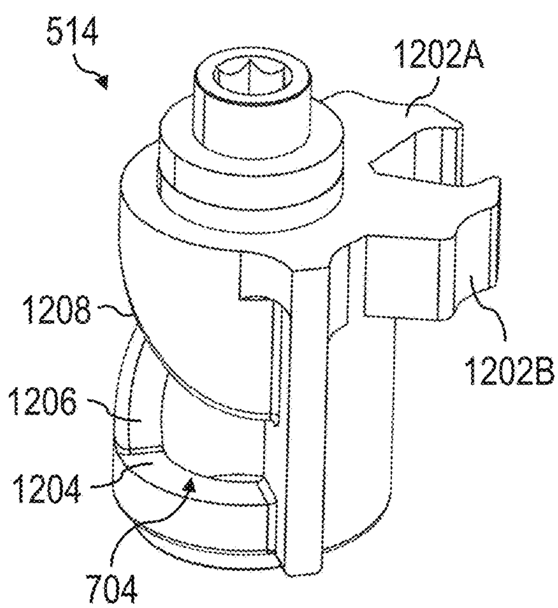

FIG. 12A

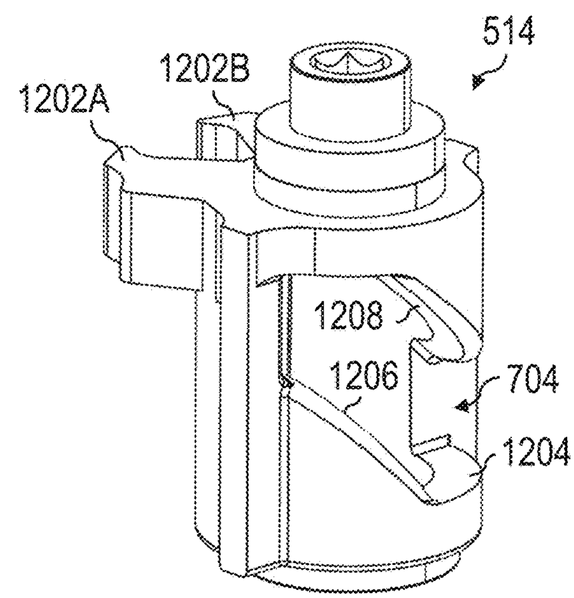

FIG. 12B

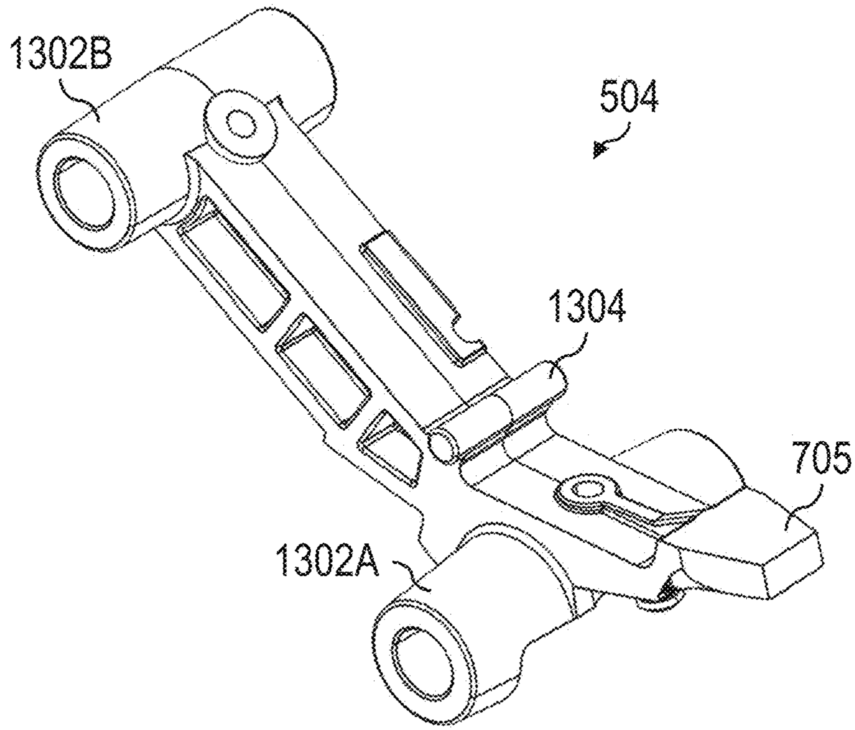

FIG. 13A

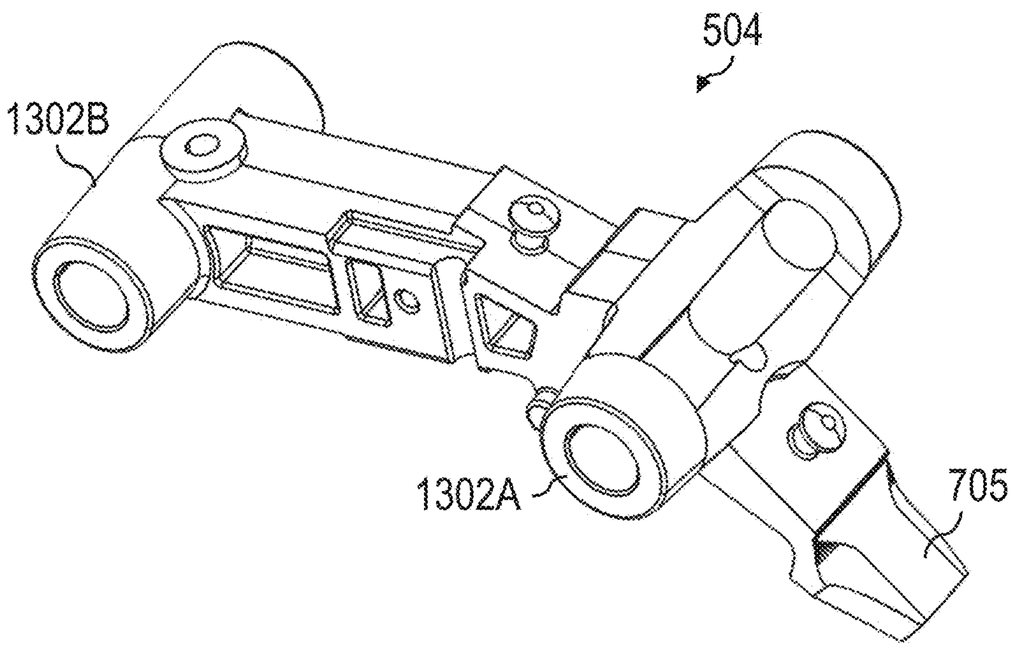

FIG. 13B

EXIT DEVICE WITH LOCKDOWN FUNCTION

RELATED APPLICATION

[0001] This Application is a Non-Provisional Application of U.S. Application Ser. No. 63/557,363, filed Feb. 23, 2024, entitled “EXIT DEVICE WITH LOCKDOWN FUNCTION”.

FIELD

[0002] Disclosed embodiments are related to lockdown exit devices.

BACKGROUND

[0003] Exit devices often include actuators such as push bars which a user may actuate to open a door. The actuators can be used to operate a latch. In some cases, the actuators may be maintained in a position. An indicator may be used to indicate a state of the exit device.

SUMMARY

[0004] In some embodiments, an exit device comprises: a latch movable between an extended latch position and a retracted latch position; a push bar movable between a first push bar position and a second push bar position, wherein movement of the push bar from the first push bar position to the second push bar position moves the latch from the extended latch position to the retracted latch position; and a lockdown assembly configured to transition between a non-lockdown configuration and a lockdown configuration, wherein in the lockdown configuration the lockdown assembly is configured to maintain the push bar in the first push bar position.

[0005] In some embodiments, an exit device comprises: a latch movable between an extended latch position and a retracted latch position; a push bar movable between a first push bar position and a second push bar position, wherein movement of the push bar from the first push bar position to the second push bar position moves the latch from the extended latch position to the retracted latch position; a lockdown assembly configured to transition between a non-lockdown configuration and a lockdown configuration, wherein in the lockdown configuration the lockdown assembly is configured to maintain the push bar in the first push bar position; and an indicator configured to indicate the exit device is in a lockdown configuration in response to the lockdown assembly transitioning to the lockdown configuration.

[0006] It should be appreciated that the foregoing concepts, and additional concepts discussed below, may be arranged in any suitable combination, as the present disclosure is not limited in this respect. Further, other advantages and novel features of the present disclosure will become apparent from the following detailed description of various non-limiting embodiments when considered in conjunction with the accompanying figures.

BRIEF DESCRIPTION OF DRAWINGS

[0007] The accompanying drawings are not intended to be drawn to scale. In the drawings, each identical or nearly identical component that is illustrated in various figures may be represented by a like numeral. For purposes of clarity, not every component may be labeled in every drawing. In the drawings:

[0008] FIG. **1**A shows a schematic view of an exit device configured to be coupled to a door including multiple rods according to some embodiments;

[0009] FIG. **1**B shows a schematic view of an exit device configured to be coupled to a door including one rod according to some embodiments;

[0010] FIG. **2** shows a schematic view of an exit device configured to be coupled to a door including a latch assembly according to some embodiments;

[0011] FIG. **3** shows a perspective view of an exit device according to some embodiments;

[0012] FIG. **4**A shows a perspective view of an exit device with an indicator providing a first indication according to some embodiments;

[0013] FIG. **4**B shows a perspective view of the exit device with the indicator providing a second indication according to some embodiments;

[0014] FIG. **5**A shows a perspective view of an exit device in a lockdown configuration without a chassis according to some embodiments;

[0015] FIG. **5**B shows a perspective view of the exit device in a non-lockdown configuration without the chassis according to some embodiments;

[0016] FIG. **6**A shows a side view of the exit device in the lockdown configuration according to some embodiments;

[0017] FIG. **6**B shows a side view of the exit device in the non-lockdown configuration according to some embodiments;

[0018] FIG. **7**A shows a perspective view of a first portion of the exit device in the lockdown configuration according to some embodiments;

[0019] FIG. **7**B shows a perspective view of the first portion of the exit device in the non-lockdown configuration according to some embodiments;

[0020] FIG. **8**A shows a perspective view of a second portion of the exit device in the lockdown configuration according to some embodiments;

[0021] FIG. **8**B shows a perspective view of the second portion of the exit device in the non-lockdown configuration according to some embodiments;

[0022] FIG. **8**C shows a perspective view of the second portion of the exit device in the non-lockdown configuration including an actuator bracket according to some embodiments;

[0023] FIG. **9**A shows a perspective view of an exit device in the lockdown configuration including an indicator according to some embodiments;

[0024] FIG. **9**B shows a perspective view of the exit device in the lockdown configuration including the indicator in the first state without an indicator housing according to some embodiments;

[0025] FIG. **10**A shows a perspective view of the exit device including the indicator in the non-lockdown configuration according to some embodiments;

[0026] FIG. **10**B shows a perspective view of the exit device including the indicator in the non-lockdown configuration without an indicator housing according to some embodiments;

[0027] FIG. **11** shows a perspective view of an indicator lever arm according to some embodiments;

[0028] FIG. **12**A shows a perspective view of a drive component according to some embodiments;

[0029] FIG. **12**B shows another perspective view of the drive component according to some embodiments;

[0030] FIG. **13**A shows a perspective view of a lift arm according to some embodiments; and

[0031] FIG. **13**B shows another perspective view of the lift arm according to some embodiments.

DETAILED DESCRIPTION

[0032] Doors are often used to selectively restrict access to a space, such as an interior of a room or building. Doors often include exit devices that selectively lock and unlock the door to selectively restrict access to the space. The exit devices can include an indicator configured to indicate whether the door is locked or unlocked. Some exit devices can be transitioned from a non-lockdown state to a lockdown state. The inventors have identified a need for an exit device which may be quickly transitioned to a lockdown state. Examples of environments where an exit device that is quickly transitioned to a lockdown state may be beneficial include classrooms, bathrooms, offices, meeting rooms, and any other suitable room or space where privacy, shelter, or restricted access are desired. This may be useful in instances where restricting access to the interior of a room is desired. In such instances, a latch actuator such as a door knob or handle on an exterior side of the door is disabled, thereby preventing operation of the exit device on the exterior side while allowing operation of the exit device using an interior latch actuator such as a push bar disposed on an interior side of the door. This may serve to prevent to prevent intruders or other persons intended from entering the interior of the room. As such, the inventors have recognized an exit device that may be quickly transitioned to a lockdown state. The exit device may provide other benefits such as remote operation of the exit device and/or quickly providing an indication of the state of the exit device.

[0033] An exit device may include a latch movable between an extended latch position and a retracted latch position. The exit device may also include a push bar movable between a first push bar position and a second push bar position. Movement of the push bar from the first push bar position to the second push bar position may move the latch from the extended latch position to the retracted latch position. The exit device may include a lockdown actuator. The lockdown actuator may be actuated to transition the lockdown assembly between a non-lockdown configuration and a lockdown configuration. In the lockdown configuration, the lockdown assembly may be configured to maintain the push bar in the first push bar position. In some cases, the lockdown assembly is configured to move the push bar to the first push bar position and maintain the push bar in first push bar position.

[0034] In some cases, the exit device includes a lockdown interface such as a rotatable lock cylinder. The lock cylinder is rotatable using a key in some embodiments. In some cases, the lock cylinder is rotatable via a thumb turn or knob. Rotating the lock cylinder may transition the lockdown assembly between the non-lockdown configuration and the lockdown configuration. For example, in response to rotation of the lock cylinder, the lockdown actuator may be configured to transition the lockdown assembly between the non-lockdown configuration and the lockdown configuration. The lockdown actuator may be formed as a motor, linear actuator, or any other appropriate mechanical or electromechanical actuator.

[0035] In some embodiments, the lockdown actuator may be operatively coupled to a lift arm and may be configured to move the lift arm between a first lift arm position and a second lift arm position. The lift arm may be movably coupled to the push bar. As such, the lockdown actuator may be configured to move the push bar via the lift arm. For example, the lockdown actuator may be controlled to move and/or maintain the push bar in the first push bar position, thereby preventing operation of the push bar and restricting access to an associated door and space. In some cases one or more components may be used to couple movement of the lockdown actuator to the lift arm. For example, the lockdown actuator may be coupled to the lift arm via a primary connecting member. In some embodiments, the primary connecting member is movably coupled to a secondary connecting member, and the secondary connecting member is movably coupled to the lift arm. In further embodiments, the secondary connecting member is movably coupled to a tertiary connecting member, and the tertiary connecting member is movably coupled to the lift arm. Any appropriate movable component of the exit device may be biased in a direction using one or more biasing elements. For example, the primary connecting member may be movable between a first primary connecting member position and a second primary connecting member position and may be biased towards the first primary connecting member position using a primary connecting member biasing element. In another example, the push bar may be biased to the first push bar position using one or more push bar biasing elements.

[0036] In some embodiments, the exit device includes an indicator configured to indicate a state of the exit device. For example, the indicator may be configured to indicate whether the lockdown assembly is in the non-lockdown configuration or the lockdown configuration. The indicator can include one or more indicator faces configured to provide an indication indicative of the state of the exit device. In some embodiments, the indicator face is visible through an indicator window. Optionally, one or more portions of the indicator or exit device are illuminated using a light source to provide an indication of the state of the exit device. In some cases, the lockdown assembly is able to be controlled remotely. The lockdown assembly may also be operated based on a predetermined schedule. For example, during a first period of time (e.g., business hours) the lockdown assembly may be configured to be in the non-lockdown configuration and during a second period of time the lockdown assembly may be configured to be in the lockdown configuration. In some cases, the exit device is operable using an access card (e.g., key card). In such cases, the override assembly may optionally not be operable via the access card. In some embodiments, the lockdown assembly is operable via both sides of an associated door. In some cases, the lockdown assembly is operable via a single side (e.g., the interior side) of the associated door, which may include the push bar.

[0037] The indicator may include an indicator body movable between a first indicator body position and a second indicator body position. In some cases, the indicator is configured to indicate that the exit device is in a lockdown state while the indicator body is in the first indicator body position and indicate that the exit device is in a non-lockdown state while the indicator body is in the second indicator body position. The indicator may include one or more indicator faces, which may be configured to provide the indication indicative of the state of the exit device. For example, the indicator may include a first indicator face configured to indicate the exit device is in a lockdown state and a second indicator face configured to indicate the exit device is in a non-lockdown state. In embodiments including an indicator window, one of the first indicator face and the second indicator face may be visible through the indicator window. In some cases, the indicator includes an indicator body biasing element configured to bias and/or apply a force to the indicator body.

[0038] Turning to the figures, specific non-limiting embodiments are described in further detail. It should be understood that the various systems, components, features, and methods described relative to these embodiments may be used either individually and/or in any desired combination as the disclosure is not limited to only the specific embodiments described herein.

[0039] As discussed herein, the exit device is configured to be coupled to a door according to some embodiments. For example, the exit device is shown coupled to a door in the depicted embodiments of FIGS. **1-2**. More specifically, a door system **10** includes an exit device **100** having a push bar **110**. In the depicted embodiments of FIGS. **1**A and **1**B, the exit device **100** also includes one or more rods **120** for securing the door. The rod **120** may be operatively coupled to a latch and/or latch assembly, which may be used to secure the door. For example, the push bar **110** may be actuated, and the rod **120** may move a latch or latch assembly to secure the door. Depending on the embodiment, a single rod or multiple rods are used. For example, in the depicted embodiment of FIG. **1**A, the exit device **100** includes two rods **120**A, **120**B. In the depicted embodiment of FIG. **1**A, the rods **120**A, and **120**B are configured to secure the door on both of an upper and lower portion of a door frame. In another example, such as the depicted embodiment of FIG. **1**B, the exit device **100** includes a single rod **120** configured to secure the door on an upper portion of a door frame. In the depicted embodiment of FIG. **2**, the exit device includes a latch assembly **130** configured to secure the door on a side portion of the door frame. The latch assembly may include a door latch (e.g., latch or bolt) configured to secure the door via the side portion of the door frame. The exit device **100** may also include a lockdown assembly **112** having a lockdown interface **113** and an indicator **114** having an indicator window **115**, as described herein and further below. In the depicted embodiments of FIGS. **1-2**, the lockdown interface **113** and indicator window **115** are disposed on a face of a chassis **102** of the exit device **100**. It should be understood that the exit device **100** may optionally be formed with the indicator **114**. Accordingly, the embodiments described herein, including the depicted embodiments of FIGS. **1**A, **1**B, and **2** may be formed without the indicator **114**. It should be appreciated that the indicator **114**, indicator window **115**, lockdown assembly **112**, and lockdown interface **113** may be positioned on any appropriate portion of the chassis **102** or other portion of the exit device **100** as the disclosure is not limited in this sense.

[0040] FIG. **3** shows a perspective view of the exit device **100** according to some embodiments. The chassis **102** may include a front chassis phase **304**. In the depicted embodiment of FIG. **3**, the lockdown interface **113** and the indicator window **115** are formed in the front chassis plate **304**. The depicted embodiment of FIG. **3** also shows a key **302** engaged with the lockdown interface **113**. FIGS. **4**A and **4**B show perspective views of the exit device **100** in a non-lockdown state, and a lockdown state, respectively. In the depicted embodiment of FIG. **4**A, the exit device **100** is in a non-lockdown state, and the indicator **114** provides an indication indicative of the exit device **100** being in the non-lockdown state. Specifically, an indicator face indicative of the non-lockdown configuration is visible through the indicator window **115**. In the depicted embodiment of FIG. **4**A, the push bar **110** is shown in the second push bar position. In the depicted embodiment of FIG. **4**B, the exit device **100** is in the lockdown state, and the indicator **114** provides an indication indicative of the exit device **500** being in the lockdown state. Specifically, in indicator face indicative of the exit device **100** being in the lockdown state is visible through the indicator window **115**. In the depicted embodiment of FIG. **4**B, the push bar **110** is shown in the first push bar position. As previously described, the push bar **110** may be maintained in the first push bar position while the exit device **100** is in the lockdown configuration.

[0041] While embodiments of the indicator faces are depicted in the figures, it should be appreciated that the indicator face may include any appropriate text, logos, icons, shapes, colors, numbers, or other visual indications as the disclosure is not limited in this sense. In some cases, the indicator face configured to indicate the non-lockdown configuration may be formed to be blank (e.g., not containing any markings) to be non-descript, and the indicator face configured to indicate the lockdown configuration may be formed to include markings or other indications to more easily indicate to the user that the exit device is in the lockdown configuration.

[0042] The depicted embodiments of FIGS. **5**A and **5**B show perspective views of the exit device **100** in the lockdown configuration and the non-lockdown configuration, respectively. The exit device **100** may include one or more lift members **502**. For example, the depicted embodiments of FIGS. **5**A and **5**B include a first lift member **502**A and a second lift member **502**B. The lift member **502** may be coupled to the push bar **110** and may be configured to move in conjunction with the push bar **110**. The lift member **502** may be movable between a first lift member position and a second lift member position. In embodiments where the push bar is configured to move in conjunction with the lift member **502**, the push bar may move between the first push bar position and the second push bar position in conjunction with the lift member **502** moving between the first lift member position and the second lift member position. In the depicted embodiments of FIGS. **5**A and **5**B, the lift member **502** is formed as a bracket, however the lift member **502** may be formed in any appropriate geometry as the disclosure is not limited in this sense. Further, the exit device **100** may include any appropriate quantity of lift members is a disclosure is also not limited in this sense.

[0043] The lift member **502** may be movably coupled to the lift arm **504**. The lift arm **504** may be movable between a first lift arm position and a second lift arm position. In some cases, the lift arm **504** may be directly and movably coupled to the push bar **110**. The exit device **100** may include any appropriate quantity of lift arms **504**, for example, in the depicted embodiment of FIGS. **5**A and **5**B, the exit device **100** includes a first lift arm **504**A and a second lift arm **504**B. In some cases, the lift arm **502** is configured to rotate between the first lift arm position and the second lift arm position. Movement of the lift arm from the first lift arm position to the second lift arm position may move the lift member **502** from the first lift member position to the second lift member position. Similarly, movement of the push bar **110** and or the lift member **502** from the first push bar position to the second push bar position and/or the lift member from the first lift member position to the second lift member position may move the lift arm **504** from the first lift arm position to the second lift arm position.

[0044] The first lift arm **504**A may be operatively coupled to a lockdown actuator **512**. For example, lockdown actuator **512** may be configured to move the lift arm **504**A between the first lift arm position and the second lift arm position. In the depicted embodiments of FIGS. **5**A and **5**B, the lockdown actuator **512** is depicted as a motor. In some cases, the lockdown actuator **512** is configured to move the lift arm **504**A to the first lift arm position in response to power being shut off to the lockdown actuator **512**. As such, in response to power being shut off from the motor, the push bar **110** may move to the first push bar position and the latch may move to the extended latch position. Similarly, the lockdown actuator **512** may be configured to move the lift arm **504** from the first lift arm position to the second lift arm position in response to power being provided to the lockdown actuator **512**. The lockdown actuator **512** may be operatively coupled to and controlled by a controller **524**. The controller **524** may also be operatively coupled to the lockdown interface **113**. In some cases, the controller **524** is configured to control the lockdown output actuator **512** in response to actuation of the lockdown interface **113**.

[0045] In some embodiments, the lockdown actuator **512** is configured to move the lift arm **504** via one or more intervening components. For example, the exit device **100** may include a primary connecting member **506** movable between a first primary connecting member position and a second primary connecting member position. The lockdown actuator **512** may be configured to move the primary connecting member **506** between the first primary connecting member position and the second primary connecting member position. Exit device **100** may also include a secondary connecting member **508** configured to move between a first secondary connecting member position and a second secondary connecting member position. In some cases, the secondary connecting member **508** may be configured to rotate between the first secondary connecting member position and the second secondary connecting member position. The secondary connecting member **508** may be movably coupled to the primary connecting member **506**, such that movement of the primary connecting member from the first primary connecting member position to the second primary connecting member position moves the secondary connecting member from the first secondary connecting member position to the second secondary connecting member position. Exit device **100** may also include a tertiary connecting member **510** movable between a first tertiary connecting member position and a second tertiary member connecting position. In some cases, the tertiary connecting member **510** is configured to rotate between the first tertiary connecting member position and the second tertiary connecting member position. In the depicted embodiments of FIGS. **5**A and **5**B, the primary connecting member **506** is movably coupled to the lockdown actuator **512** and the secondary connecting member **508**. In the depicted embodiments of FIGS. **5**A and **5**B, the secondary connecting member **508** is movably coupled to the primary connecting member **506** and the tertiary connecting member **510**. In the depicted embodiments of FIGS. **5**A and **5**B, the tertiary connecting member **510** is movably coupled to the secondary connecting member **508** and the first lift arm **504**A.

[0046] The exit device **100** may also include the second lift arm **504**B which may be movably coupled to the second lift member **502**B. The second lift arm five of **4**B may also be movably coupled to a drive component **514**. The drive component may be configured to move between a first drive component position and a second drive component position. In some cases, the drive component rotates between the first drive component position and the second drive component position. The drive component may be configured to move from the first tried component position to the second drive component position in response to the lift arm **504**B moving from the first lift arm position to the second arm position. It should be understood that movement of the push bar **110** (not shown in FIGS. **5**A and **5**B) from the first push bar position to the second push bar position may move the lift arms **504**A, **504**B and/or the lift members **502**A, **502**B, from their respective first lift arm positions to the second lift arm positions and/or first lift member positions to the second lift member positions.

[0047] The depicted embodiment of the exit device **100** in figs five a and **5**B includes an indicator **114**. The indicator **114** may include an indicator housing **520** configured to support components of the indicator **114**. Indicator **114** may also include an indicator biasing element **522** configured to bias and/or apply a force to one or more portions of the indicator **114**, discussed further with respect to proceeding figures. The exit device **100** may include an indicator lever arm **513** movably coupled to the drive component **514** and configured to move between a first indicator lever arm position and a second indicator lever arm position. In some cases, the indicator lever arm **513** is configured to rotate between the first indicator lever arm position and the second indicator lever arm position.

[0048] In the depicted embodiment of FIG. **5**A, the lift members **502**A, **502**B are shown in the first lift member position, the lift arms **504**A, **504**B are shown in the first lift arm position, the primary connecting member **506** is shown in first primary connecting member position, the secondary connecting member **508** is shown in the first secondary connecting member position, the tertiary connecting member **510** is shown in the first tertiary connecting member position, the drive component **514** is shown in the first drive component position, and the indicator lever arm **513** is shown in the first indicator lever arm position.

[0049] In the depicted embodiment of FIG. **5**B, the lift members **502**A, **502**B are shown in the second lift member position, the lift arms **504**A, **504**B are shown in the second lift arm position, the primary connecting member **506** is shown in second primary connecting member position, the secondary connecting member **508** is shown in the second secondary connecting member position, the tertiary connecting member **510** is shown in the second tertiary connecting member position, the drive component **514** is shown in the second drive component position, and the indicator lever arm **513** is shown in the second indicator lever arm position.

[0050] FIGS. **6**A and **6**B show side views the exit device **100** in a lockdown state and a non-lockdown state, respectively, according to some embodiments. In the depicted embodiment of FIG. **6**A, the push bar **110** is shown in the first push bar position. The depicted embodiment of FIG. **6**B, the push bar **110** is shown in the second push bar position. The exit device **100** may include an actuator bracket **602**, which is described further with respect to proceeding figures. The exit device **100** may also include a bottom exit device housing **604**, which may be configured to support one or more components of the exit device **100**.

[0051] FIGS. **7**A and **7**B depict a portion of the exit device **100** in a lockdown state and a non-lockdown state, respectively. As previously discussed, the lockdown interface **113** may be used to transition the exit device **100** between the lockdown state and a non-lockdown state. In the depicted embodiments of FIGS. **7**A and **7**B, the lockdown interface **113** is formed as a lock cylinder configured to receive the key **302**. A cam plate **702** may be movably coupled to the lockdown interface **113** and movable between a first cam plate position and a second cam plate position. For example, rotation of the lockdown interface **113** from a first lockdown interface position to a second lockdown interface position may rotate the cam plate **702** from the first cam plate position to the second cam plate position. The cam plate **702** may be configured to engage with a lockdown switch **708**, and/or the lockdown switch **708** may be configured to detect a position and/or configuration of the cam plate **702** and/or lockdown interface **113**. The lockdown switch **708** may be operatively coupled to the controller **524**, and the controller **524** may be configured to obtain the detected position and/or configuration from the lockdown switch **708**. In some cases, the cam plate **702** may be configured to contact and change the state of the lockdown switch **708** as the cam plate **702** moves.

[0052] As previously discussed, the lift arm **504** may be configured to engage the drive component **514**. The drive component **514** may include a drive channel **704** configured to receive a lift arm projection **705** of the lift arm **704**. The drive channel **704** may include one or more spiral/helical ramps, discussed further with respect to proceeding figures. Movement of the lift arm **504** from the first lift arm position to the second lift arm position may move the lift arm projection **705** within the drive channel **704**, such that the lift arm projection **705** engages a surface of the drive channel **704** while the lift arm **504** moves, thereby moving the drive component **514** from the first drive component position to the second drive component position. Movement of the drive component **514** from the first drive component position to the second drive component position may cause one or more portions of the drive component **514** to engage one or more portions of the indicator lever arm **513**, discussed further with respect to proceeding figures. For example, movement of the drive component **514** from the first drive component position to the second drive component position may move the indicator lever arm **513** from the first indicator lever arm position to the second indicator lever arm position. As will be discussed further with respect to proceeding figures, movement of the indicator lever arm **513** may move a portion of the indicator **114**, thereby changing the indication provided by the indicator **114**. The exit device **100** may include a lift arm biasing element **706**B configured to bias the lift arm **504**B and/or the lift member **502**B, which optionally may be formed as a torsion spring. For example, the lift arm biasing element **706**B may be configured to bias the lift arm **504**B and/or the lift member **502**B towards the first lift arm position and/or the first lift member position.

[0053] FIGS. **8**A and **8**B show another portion of the exit device **100** in a lockdown state and a non-lockdown state, respectively. FIG. **8**C shows the portion of the exit device **100** including an actuator bracket **602** in the non-lockdown state. In some embodiments, the exit device **100** includes a connecting member biasing element **802** configured to bias one or more of the lockdown actuator **512**, the first connecting member **506**, the second connecting member **508**, the tertiary connecting member **510**, and the lift arm **504**A in a direction. For example, in the depicted embodiments of FIGS. **8**A, **8**B, and **8**C, the connecting member biasing element **802** is coupled to the lockdown actuator **512** and the primary connecting member **506** and is configured to bias the primary connecting member **506** to the first primary connecting member position. However, it should be appreciated that the connecting member biasing element **802** may be coupled to any appropriate components of the exit device **100** as the disclosure is not limited to the depicted embodiments.

[0054] In the depicted embodiments of FIGS. **8**A, **8**B, and **8**C, the primary connecting member **506** is configured to move linearly between the first primary connecting member position and the second primary connecting member position. Optionally, the primary connecting member **506** may be slidingly coupled to the actuator bracket **602**. For example, a portion of the primary connecting member **506** may be configured to slide within an actuator bracket slot **804** of the actuator bracket **602**. In the depicted embodiments of FIGS. **8**A, **8**B, and **8**C, the secondary connecting member **508** is configured to rotate between the first secondary connecting member position and the second secondary connecting member position. Also in the depicted embodiments, the secondary connecting member **508** includes a slot configured to receive a portion of the tertiary connecting member **510**. It should be appreciated that each of the primary connecting member **506**, the secondary connecting member **508**, and the tertiary connecting member **510** may be formed in any appropriate geometry as the disclosure is not limited in this sense.

[0055] In the depicted embodiments of FIGS. **8**A, **8**B, and **8**C, the lockdown actuator **512** is formed as a motor configured to selectively rotate a threaded shaft. The threaded shaft may be configured to move the first connecting member **506** linearly upon rotation. In some embodiments, the threaded shaft retracts into the motor upon power being shut off to the motor. In the depicted embodiments of FIGS. **8**A, **8**B, and **8**C, the lift member **502**A is biased to the first lift member position using a lift arm biasing member **706**A.

[0056] FIG. **9**A shows a portion of the exit device **100** in the lockdown state according to some embodiments. FIG. **9**B also shows the portion of the exit device **100** in the lockdown state, without the indicator housing **520** shown according to some embodiments. As best seen in the depicted embodiment of FIG. **9**B, the indicator **114** may include an indicator body **902**. The indicator body **902** may be movable between a first indicator body position and a second indicator body position. One or more indicator faces **904** may be coupled to and configured to move with the indicator body **902**. For example, in the depicted embodiment of FIG. **9**B, the indicator **114** includes two indicator faces **904**A, **904**B. The indicator face **904**A may be associated with and configured to indicate the lockdown configuration. The indicator face **904**B may be associated with him configured to indicate the non-lockdown configuration. The indicator body **902** may be configured to rotate between the first indicator body position and the second indicator by position. The indicator lever arm **513** may be configured to move the indicator body **902** between the first indicator body position and the second indicator body position. For example, movement of the indicator lever arm **513** from the first indicator lever arm position to the second indicator lever arm position may move the indicator body **902** from the first indicator body position to the second indicator body position. Similarly, movement of the indicator lever arm **513** from the second indicator lever arm position to the first indicator lever arm position may move the indicator body **902** from the second indicator body position to the first indicator body position. One or more portions of the indicator body **902**, such as one or more projections of the indicator body may be configured to engage the indicator lever arm **513** to couple movement of the indicator lever arm **513** to the indicator body **902**. In the depicted embodiments of FIGS. **9**A and **9**B the indicator body **902** is shown in the first indicator body position and the indicator lever arm **513** is shown in the first indicator lever arm position.

[0057] The depicted embodiments of FIGS. **10**A and **10**B show the same portion of the exit device as shown in FIGS. **9**A and **9**B, with the exit device in the non-lockdown configuration. In the depicted embodiments of FIGS. **10**A and **10**B, the indicator body **902** is shown in the second indicator body position in the indicator lever arm **513** shown in the second indicator lever arm position. As previously discussed, the indicator **114** may include an indicator biasing element **522**. The indicator biasing element **522** may be coupled to one are more portions of the indicator body **902** and/or the indicator body **520**. In some cases, the indicator biasing element **522** may be configured to maintain a tension in association with the indicator body **902** to assist in providing stable movement of the indicator body **902**. In some cases, indicator biasing element **522** may be configured to bias the indicator body **902** to either the first indicator body position or the second indicator body position.

[0058] FIG. **11** shows a perspective view of the indicator lever arm **513** according to some embodiments. The indicator lever arm **513** may include an indicator lever body **1102** with indicator lever projection **1104** coupled thereto. One or more indicator lever prongs **1106**A, **1106**B may also be coupled to the indicator lever body **1102**. The indicator lever prongs **1106**A, **1106**B may be formed proximate to the indicator lever projection **1104**. In some cases, the indicator lever arm **513** may be configured to rotate about an indicator lever hub **1110**, which is coupled to the indicator body **1102**. The indicator lever arm **513** may also include an indicator lever tab **1108** which may be coupled to one or both of the indicator lever hub **1110** and the indicator lever body **1102**. The indicator lever projection **1104** may be configured to engage one or more portions of the drive component **514**. Optionally, the indicator lever prongs **1106**A, **1106**B may also be configured to engage one or more portions of the drive component **514**. The indicator lever tab **1108** may be configured to engage the indicator body **902**, as described herein. It should be understood that the indicator lever arm **513** may be formed in any appropriate geometry as the disclosure is not limited to the depicted embodiment of FIG. **11**.

[0059] FIGS. **12**A and **12**B depict the drive component **514** according to some embodiments. The drive component **514** may include one or more portions configured to engage the indicator lever arm **513** as described herein. For example, the drive component **514** may include one or more drive tabs **1202**A, **1202**B configured to engage one or more of the indicator lever projection **1104** and the indicator lever prongs **1106**A, **1106**B. For example, rotation of the drive component **514** may cause one or both of the drive tabs **1202**A, **1202**B to engage the indicator lever projection **1104** to rotate the indicator lever arm **513**.

[0060] The drive component **514** may include a drive channel **704** as previously discussed. The drive channel **704** may include a bottom drive surface **1204**, a bottom drive ramp **1206**, and a top drive ramp **1208**. In some embodiments, the bottom drive ramp **1206** and the top drive ramp **1208** form a helical/spiral geometry. Also previously discussed, movement of the lift arm **504** may move the drive component **514**. For example, the lift arm projection **705** may be configured to engage the bottom drive ramp **1206** and the top drive ramp **1208**. Movement of the lift arm **504** between the first lift arm position and the second lift arm position may cause the lift arm projection **705** to engage the bottom drive ramp **1206** or the top drive ramp **1208**, depending on the direction of movement of the lift arm **504**. For example, movement of the lift arm **504** from the first lift arm position to the second lift arm position may cause the lift arm projection **705** to engage the top drive ramp **1208**, causing the drive component **514** to rotate from the first drive component position to the second drive component position. Similarly, movement of the lift arm **504** from the second lift arm position to the first lift arm position may cause the lift arm projection **705** to engage the bottom drive ramp **1206**, causing the drive component **514** to rotate from the second drive component position to the first drive component position.

[0061] FIGS. **13**A and **13**B depict the lift arm **504** according to some embodiments. The lift arm **504** may include one or more lift arm hubs **1302**A, **1302**B. Optionally, the lift arm **504** may be configured to rotate about one or both of the lift arm hubs **1302**A, **1302**B. In some embodiments, the lift arm **504** may include a lift arm catch **1304**, which may be configured to receive the lift arm biasing element **706**. It should be appreciated that the lift arm **504** may be formed in any appropriate geometry as the disclosure is not limited sense.

[0062] While the present teachings have been described in conjunction with various embodiments and examples, it is not intended that the present teachings be limited to such embodiments or examples. On the contrary, the present teachings encompass various alternatives, modifications, and equivalents, as will be appreciated by those of skill in the art. Accordingly, the foregoing description and drawings are by way of example only.

What is claimed is:

1. An exit device comprising:

a latch movable between an extended latch position and a retracted latch position;

a push bar movable between a first push bar position and a second push bar position, wherein movement of the push bar from the first push bar position to the second push bar position moves the latch from the extended latch position to the retracted latch position; and a lockdown assembly configured to transition between a non-lockdown configuration and a lockdown configuration, wherein in the lockdown configuration the lockdown assembly is configured to maintain the push bar in the first push bar position.

2. The exit device of claim **1**, wherein the lockdown assembly is further configured to maintain the latch in the latch extended position.

3. The exit device of claim **1**, further comprising a lift arm movably coupled to the push bar and movable between a first lift arm position and a second lift arm position, wherein movement of the push bar from the first push bar position to the second push bar position moves the lift arm from the first lift arm position to the second lift arm position.

4. The exit device of claim **3**, wherein the lockdown assembly comprises a lockdown actuator configured to move the lift arm from the first lift arm position to the second lift arm position.

5. The exit device of claim **4**, wherein the lockdown actuator is a motor.

6. The exit device of claim **4**, wherein the motor is configured to move a connecting member from a first connecting member position to a second connecting member position, and wherein movement of the connecting member from the first connecting member position to the second connecting member position moves the lift arm from the first lift arm position to the second lift arm position.

7. The exit device of claim **5**, wherein the lockdown assembly is configured to shut off power to the motor in the lockdown configuration.

8. The exit device of claim **7**, further comprising a push bar biasing element configured to bias the push bar to the first push bar position, and wherein in response to power being shut off to the motor, the push bar moves to the first push bar position.

9. The exit device of claim **1**, wherein the lockdown assembly includes a rotatable lock cylinder, wherein rotation of the lock cylinder transitions the lockdown assembly between the non-lockdown configuration and the lockdown configuration.

10. The exit device of claim **9**, wherein the lockdown assembly includes a cam plate rotatably coupled to the lock cylinder, wherein the cam plate is configured to rotate in conjunction with the lock cylinder to activate a switch, wherein activation of the switch transitions the lockdown assembly between the non-lockdown configuration and the lockdown configuration.

11. The exit device of claim **1**, further comprising an indicator configured to indicate a state of the exit device.

12. The exit device of claim **1**, wherein the lockdown assembly is configured to be controlled to transition between the non-lockdown configuration and the lockdown configuration remotely.

13. An exit device comprising:

a latch movable between an extended latch position and a retracted latch position;

a push bar movable between a first push bar position and a second push bar position, wherein movement of the push bar from the first push bar position to the second push bar position moves the latch from the extended latch position to the retracted latch position;

a lockdown assembly configured to transition between a non-lockdown configuration and a lockdown configuration, wherein in the lockdown configuration the lockdown assembly is configured to maintain the push bar in the first push bar position; and

an indicator configured to indicate the exit device is in a lockdown configuration in response to the lockdown assembly transitioning to the lockdown configuration.

14. The exit device of claim **13**, wherein the indicator comprises an indicator body configured to move between a first indicator body position and a second indicator body position.

15. The exit device of claim **14**, further comprising a transmission component movable between a first transmission position and a second transmission position, wherein movement of the transmission component from the first transmission position to the second transmission position moves the indicator body from the first indicator body position to the second indicator body position.

16. The exit device of claim **15**, wherein the indicator further comprises an indicator lever configured to move between a first indicator lever position and a second indicator lever position, wherein movement of the indicator lever from the first indicator lever position to the second indicator lever position moves the indicator body from the first indicator body position to the second indicator body position, and wherein movement of the transmission component from the first transmission position to the second transmission position moves the lever arm from the first lever arm position to the second lever arm position.

17. The exit device of claim **15**, further comprising a lift arm movable between a first lift arm position and a second lift arm position, wherein movement of the lift arm from the first lift arm position to the second lift arm position moves the transmission component from the first transmission position to the second transmission position.

18. The exit device of claim **13**, wherein the indicator further comprises an indicator window, wherein an indication associated with the exit device is visible through the indicator window.

19. The exit device of claim **18**, wherein the indicator further comprises a first indicator face and a second indicator face, wherein the indicator is configured to display the first indicator face in response to the lockdown assembly transitioning to a non-lockdown configuration, and wherein the indicator is configured to display the second indicator face in response to the lockdown assembly transitioning to the lockdown configuration.

20. The exit device of claim **19**, wherein the first indicator face and the second indicator face are coupled to the indicator body, and wherein the indicator is configured to display the first indicator face when the indicator body is in the first indicator body position, and wherein the indicator is configured to display the second indicator face when the indicator body is in the second indicator body position.

21. The exit device of claim **20**, the indicator further comprising an indicator biasing element configured to bias the indicator body in a first direction.

22. The exit device of claim **13**, further comprising a push bar biasing element configured to bias the push bar to the first push bar position.

23. The exit device of claim **13**, further comprising a motor configured to move the push bar between the first push bar position and the second push bar position, wherein the push bar is configured to move to the first push bar position in response to power being shut off to the motor.

24. The exit device of claim **13**, further comprising a rotatable lock cylinder, wherein the lockdown assembly is configured to transition between the non-lockdown configuration and the lockdown configuration in response to rotation of the rotatable lock cylinder.

25. The exit device of claim **17**, wherein the transmission component includes a channel configured to engage the lift arm, and wherein movement of the lift arm from the first lift arm position to the second lift arm position rotates the transmission component from the first transmission component position to the second transmission component position via the channel.

26. The exit device of claim **25**, wherein the channel is formed as a helical ramp configured to engage the lift arm, and wherein movement of the lift arm from the first lift arm position to the second lift arm position rotates the transmission component from the first transmission component position to the second transmission component position via the helical ramp.

27. The exit device of claim **13**, wherein the lockdown assembly is configured to be controlled remotely.

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